

# MINI PROJECT – AIR QUALITY DATA ANALYTICS

Title:

Air Quality Data Analytics using Python, Pandas, Matplotlib, NumPy, and Exploratory Data Analysis

**Dataset:** city\_day.csv

**Records:** 29,531    **Columns:** 16

**Cities:** 26    **Date Range:** 01-01-2015 to 01-07-2020

College Practical / Mini-Project Record

# 1. Aim

To analyze the City Day air-quality dataset using Python and Exploratory Data Analysis techniques and to identify AQI distribution, city-wise differences, pollutant patterns, time trends, and relationships between pollutants and AQI.

# 2. Abstract

The project uses the supplied **city\_day.csv** dataset containing 29,531 records and 16 columns for 26 cities. The data covers the period from 01-01-2015 to 01-07-2020. Python, Pandas, NumPy and Matplotlib are used for data loading, cleaning, statistical analysis and visualization. The dataset contains 24,850 non-missing AQI values. The analysis includes AQI distribution, AQI bucket counts, city-wise average AQI, pollutant averages, time-based AQI trend, PM2.5 versus AQI, and correlation analysis. These results provide a simple data-driven view of air-quality patterns present in the supplied dataset.

# 3. Problem Statement

Air-quality data contains observations for multiple cities, dates, pollutants and AQI values. It is difficult to understand the overall condition of the data and compare cities by looking at raw rows alone. The problem is to analyze the supplied City Day dataset and present useful air-quality information through simple statistics and graphs.

# 4. Objectives

1. Load and inspect the City Day dataset.
2. Identify dataset size, data types, missing values and duplicate records.
3. Calculate basic AQI statistics such as mean, median, minimum and maximum.
4. Study the distribution of AQI buckets.
5. Compare average AQI values across cities.
6. Analyze average recorded pollutant levels.
7. Study the average AQI trend over time.
8. Measure the relationship between PM2.5 and AQI.
9. Use correlation analysis to identify relationships among pollutants and AQI.

## 5. Software Requirements

| Software / Library | Purpose                                     |
|--------------------|---------------------------------------------|
| Python             | Programming and analysis                    |
| Pandas             | Data loading, cleaning and tabular analysis |
| NumPy              | Numerical operations                        |
| Matplotlib         | Data visualization                          |
| ReportLab          | Generation of this PDF report               |

## 6. Dataset Description

The analysis is based only on the supplied **city\_day.csv** file. The dataset has 29,531 rows and 16 columns. It contains city and date information, pollutant observations, AQI and AQI bucket labels.

| Measure           | Actual value             |
|-------------------|--------------------------|
| Number of rows    | 29,531                   |
| Number of columns | 16                       |
| Number of cities  | 26                       |
| Date range        | 01-01-2015 to 01-07-2020 |
| Valid AQI count   | 24,850                   |
| Duplicate records | 0                        |
| Average AQI       | 166.46                   |
| Median AQI        | 118.00                   |
| Minimum AQI       | 13                       |
| Maximum AQI       | 2049                     |

### Column description

| Column     | Description            |
|------------|------------------------|
| City       | City name              |
| Date       | Observation date       |
| PM2.5      | PM2.5 pollutant value  |
| PM10       | PM10 pollutant value   |
| NO         | Nitric oxide value     |
| NO2        | Nitrogen dioxide value |
| NOx        | Nitrogen oxides value  |
| NH3        | Ammonia value          |
| CO         | Carbon monoxide value  |
| SO2        | Sulfur dioxide value   |
| O3         | Ozone value            |
| Benzene    | Benzene value          |
| Toluene    | Toluene value          |
| Xylene     | Xylene value           |
| AQI        | Air Quality Index      |
| AQI_Bucket | AQI category label     |

## 7. Dataset Quality Summary

| Column     | Missing values | Missing % |
|------------|----------------|-----------|
| City       | 0              | 0.00%     |
| Date       | 0              | 0.00%     |
| PM2.5      | 4,598          | 15.57%    |
| PM10       | 11,140         | 37.72%    |
| NO         | 3,582          | 12.13%    |
| NO2        | 3,585          | 12.14%    |
| NOx        | 4,185          | 14.17%    |
| NH3        | 10,328         | 34.97%    |
| CO         | 2,059          | 6.97%     |
| SO2        | 3,854          | 13.05%    |
| O3         | 4,022          | 13.62%    |
| Benzene    | 5,623          | 19.04%    |
| Toluene    | 8,041          | 27.23%    |
| Xylene     | 18,109         | 61.32%    |
| AQI        | 4,681          | 15.85%    |
| AQI_Bucket | 4,681          | 15.85%    |

The dataset contains **88,488** missing cells across all columns. AQI has 4,681 missing values, so 24,850 rows are available for AQI-based numerical analysis. No duplicate records were found using the complete-row duplicate check.

## 8. System Architecture

| Stage | Process                                                             |
|-------|---------------------------------------------------------------------|
| 1     | Read city_day.csv using Pandas                                      |
| 2     | Inspect shape, data types, missing values and duplicates            |
| 3     | Prepare date and numeric columns for analysis                       |
| 4     | Calculate descriptive statistics and grouped summaries              |
| 5     | Generate AQI, city, pollutant, trend and correlation visualizations |
| 6     | Interpret results and report findings                               |

**Flow:** Dataset → Data Loading → Data Inspection → Statistical Analysis → Visualization → Interpretation → Findings

## 9. Algorithm

**Step 1:** Start.

**Step 2:** Import Pandas, NumPy and Matplotlib.

**Step 3:** Read city\_day.csv into a Pandas DataFrame.

**Step 4:** Convert the Date column into date format.

**Step 5:** Display the first rows, shape, data types, missing values and duplicate count.

**Step 6:** Calculate AQI count, mean, median, minimum and maximum.

**Step 7:** Count the records in each AQI bucket.

**Step 8:** Calculate average AQI for each city and select the top 10 cities.

**Step 9:** Calculate average recorded value for each pollutant.

**Step 10:** Calculate average AQI by month to study the time trend.

**Step 11:** Calculate the correlation between PM2.5 and AQI.

**Step 12:** Calculate the correlation matrix for pollutants and AQI.

**Step 13:** Generate the required graphs.

**Step 14:** Interpret the graphs and prepare major findings.

**Step 15:** Stop.

## 10. Program 1 – Import Dataset

```
import pandas as pd

df = pd.read_csv("city_day.csv")
print(df.head())
```

This program imports the dataset using Pandas and displays the first five rows.

## 11. Program 2 – Dataset Information

```
print(df.info())
print(df.shape)
print(df.dtypes)
print(df.isnull().sum())
print(df.duplicated().sum())
```

This program checks the structure, size, data types, missing values and duplicate records.

## 12. Program 3 – Statistical Analysis

```
print(df.describe())
print("AQI Count:", df["AQI"].count())
print("Average AQI:", df["AQI"].mean())
print("Median AQI:", df["AQI"].median())
print("Minimum AQI:", df["AQI"].min())
print("Maximum AQI:", df["AQI"].max())
```

The statistical analysis gives the central tendency and range of the AQI values.

| AQI statistic      | Actual result |
|--------------------|---------------|
| Count              | 24,850        |
| Mean               | 166.46        |
| Median             | 118.00        |
| Minimum            | 13            |
| Maximum            | 2049          |
| Standard deviation | 140.70        |

### AQI and Bucket Analysis Code

```
plt.hist(df["AQI"].dropna(), bins=40)
plt.xlabel("AQI")
plt.ylabel("Frequency")
plt.show()

print(df["AQI_Bucket"].value_counts())
df["AQI_Bucket"].value_counts().plot(kind="bar")
plt.show()
```

### City-wise and Pollutant Analysis Code

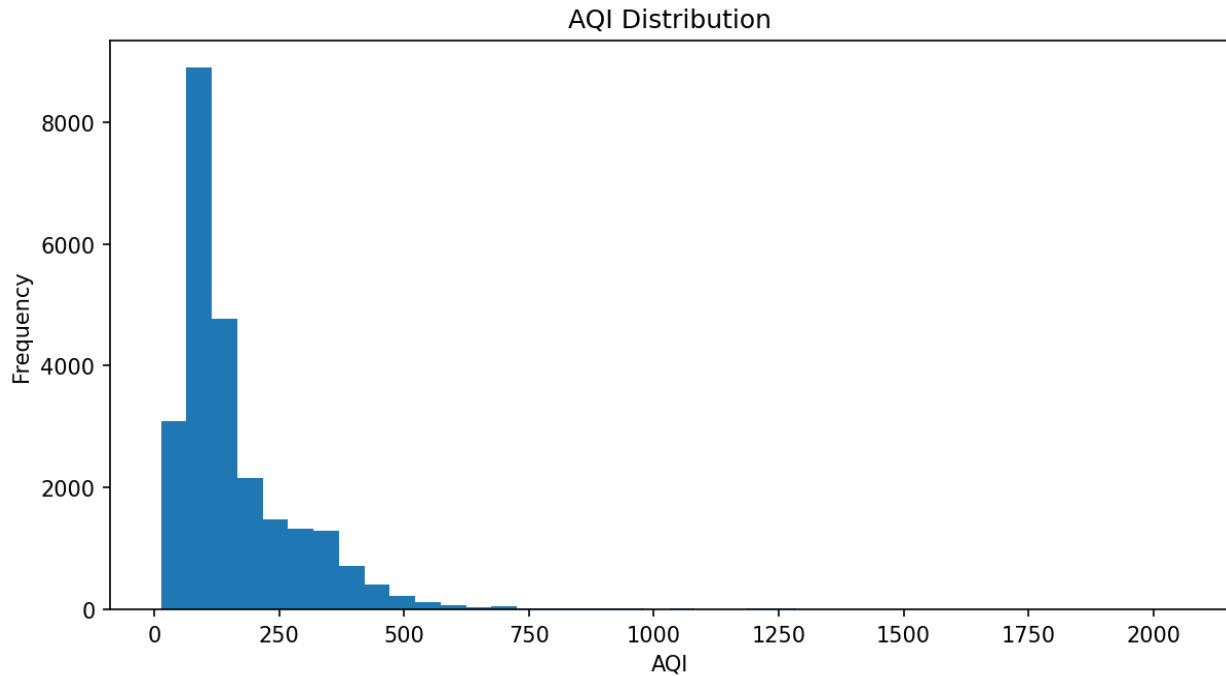
```
city_aqi = df.groupby("City")["AQI"].mean().sort_values(ascending=False)
print(city_aqi.head(10))

pollutants = ["PM2.5", "PM10", "NO", "NO2", "NOx", "NH3", "CO", "SO2", "O3", "Benzene", "Toluene", "Xylene"]
print(df[pollutants].mean().sort_values(ascending=False))
```

### 13. Analysis 1 – AQI Distribution

```
plt.hist(df["AQI"].dropna(), bins=40)
plt.xlabel("AQI")
plt.ylabel("Frequency")
plt.title("AQI Distribution")
plt.show()
```

#### Visualization



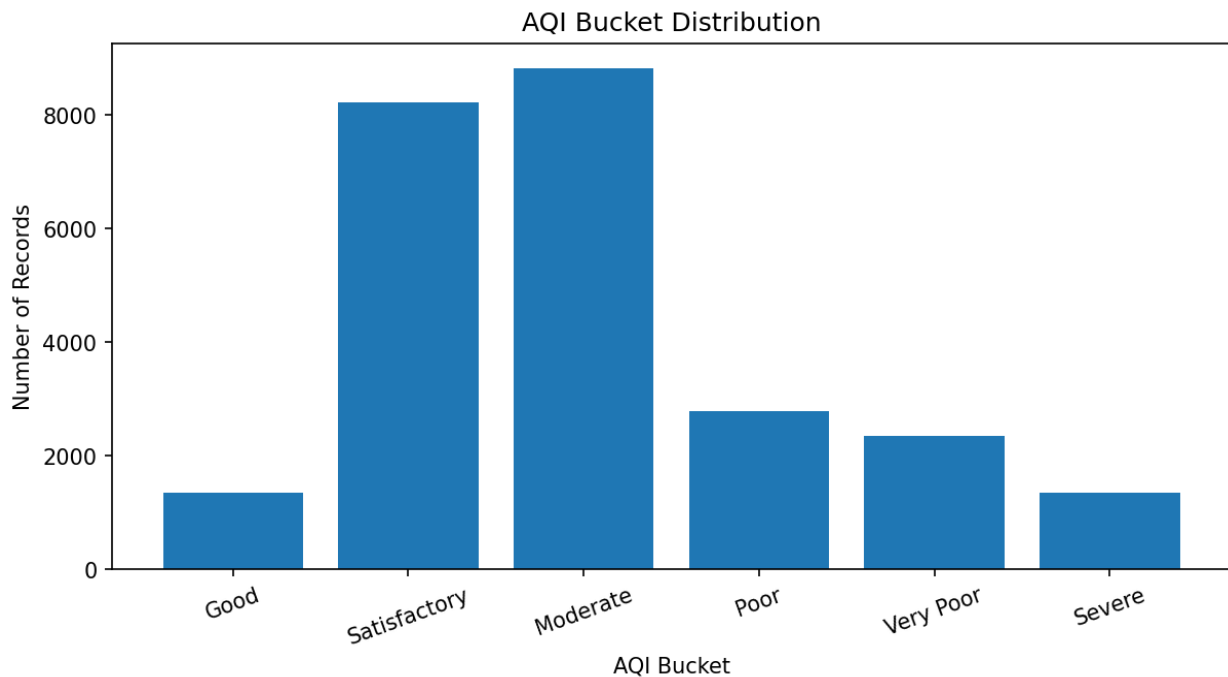
**Interpretation:** The histogram shows the distribution of the 24,850 valid AQI observations. The average AQI is 166.46 and the median is 118.00. The observed AQI range is 13 to 2049, showing a wide spread in the recorded values.

## 14. Analysis 2 – AQI Bucket Distribution

```
bucket_counts = df["AQI_Bucket"].value_counts()
print(bucket_counts)
```

```
bucket_counts.plot(kind="bar")
plt.xlabel("AQI Bucket")
plt.ylabel("Number of Records")
plt.show()
```

### Visualization



**Interpretation:** Among the labeled AQI records, **Moderate** is the most frequent category with 8,829 records. Satisfactory has 8,224 records and Poor has 2,781 records. There are 4,681 missing bucket labels.

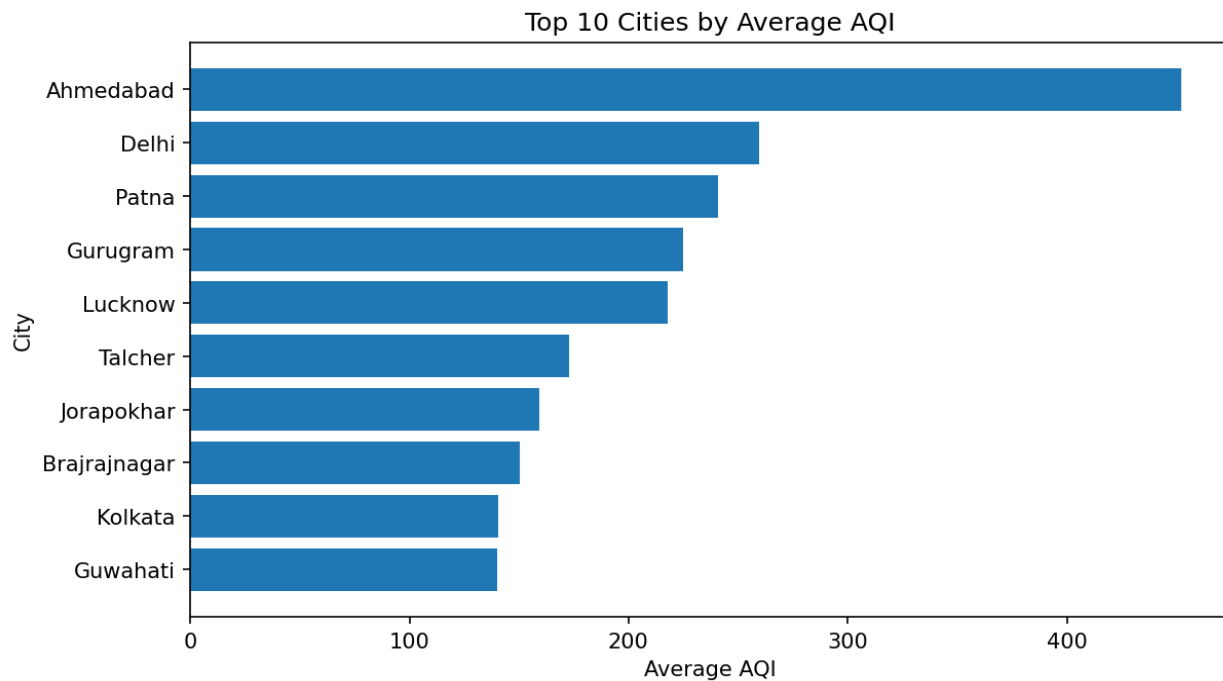
## 15. Analysis 3 – City-wise AQI Analysis

```
city_aqi = df.groupby("City")["AQI"].mean().sort_values(ascending=False)
top10 = city_aqi.head(10)
```

```
top10.plot(kind="bar")
plt.xlabel("City")
plt.ylabel("Average AQI")
plt.title("Top 10 Cities by Average AQI")
plt.xticks(rotation=45)
plt.show()
```

### Visualization





**Interpretation:** Ahmedabad has the highest average AQI at 452.12 among the cities with valid AQI observations. Delhi and Patna follow with 259.49 and 240.78, respectively. The chart highlights the difference in average AQI between the top 10 cities.

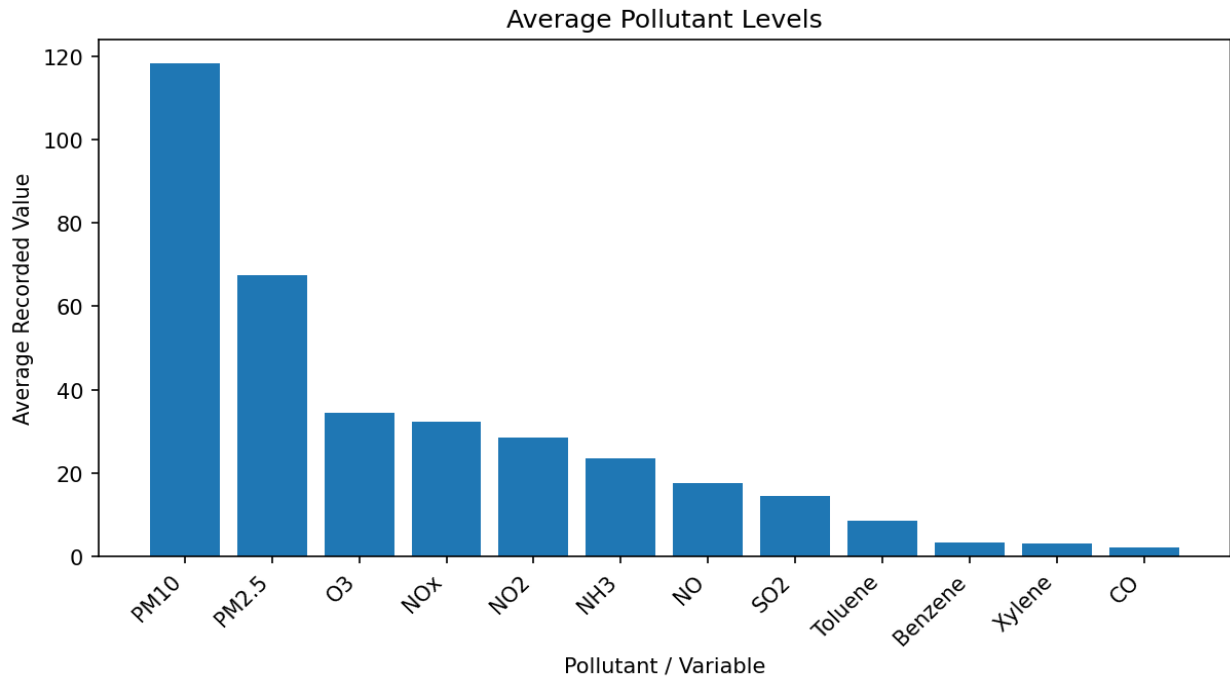
## 16. Analysis 4 – Pollutant Analysis

```
pollutants = ["PM2.5", "PM10", "NO", "NO2", "NOx", "NH3", "CO", "SO2", "O3", "Benzene", "Toluene", "Xylene"]

pollutant_avg = df[pollutants].mean().sort_values(ascending=False)
print(pollutant_avg)

pollutant_avg.plot(kind="bar")
plt.xlabel("Pollutant")
plt.ylabel("Average Recorded Value")
plt.title("Average Pollutant Levels")
plt.xticks(rotation=45)
plt.show()
```

### Visualization



**Interpretation:** PM10 has the highest average recorded value among the pollutant columns at 118.13. PM2.5 is next at 67.45, followed by O3 at 34.49. The chart compares the average recorded values in the dataset; the values are kept in the dataset's original measurement scale.

| Pollutant | Average |
|-----------|---------|
| PM2.5     | 67.451  |
| PM10      | 118.127 |
| NO        | 17.575  |
| NO2       | 28.561  |
| NOx       | 32.309  |
| NH3       | 23.483  |
| CO        | 2.249   |
| SO2       | 14.532  |
| O3        | 34.491  |
| Benzene   | 3.281   |
| Toluene   | 8.701   |
| Xylene    | 3.070   |

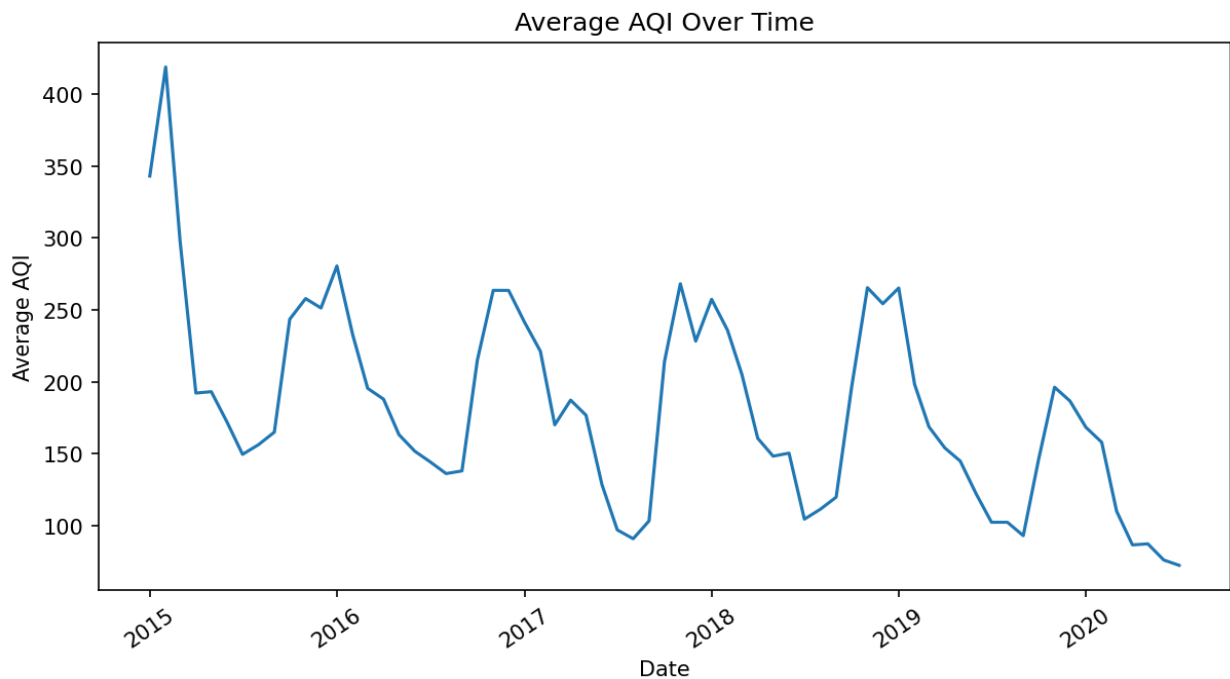
## 17. Analysis 5 – AQI Trend Over Time

```
df["Date"] = pd.to_datetime(df["Date"])

monthly_aqi = (df.set_index("Date")["AQI"].resample("MS").mean().dropna())

plt.plot(monthly_aqi.index, monthly_aqi.values)
plt.xlabel("Date")
plt.ylabel("Average AQI")
plt.title("Average AQI Over Time")
plt.show()
```

### Visualization



**Interpretation:** The line chart shows monthly average AQI from the available AQI observations. The dataset spans January 2015 to July 2020. The monthly aggregation makes longer-term changes easier to observe than individual daily records.

The highest monthly average AQI in the available monthly series was 418.83 in February 2015, while the lowest monthly average was 72.50 in July 2020.

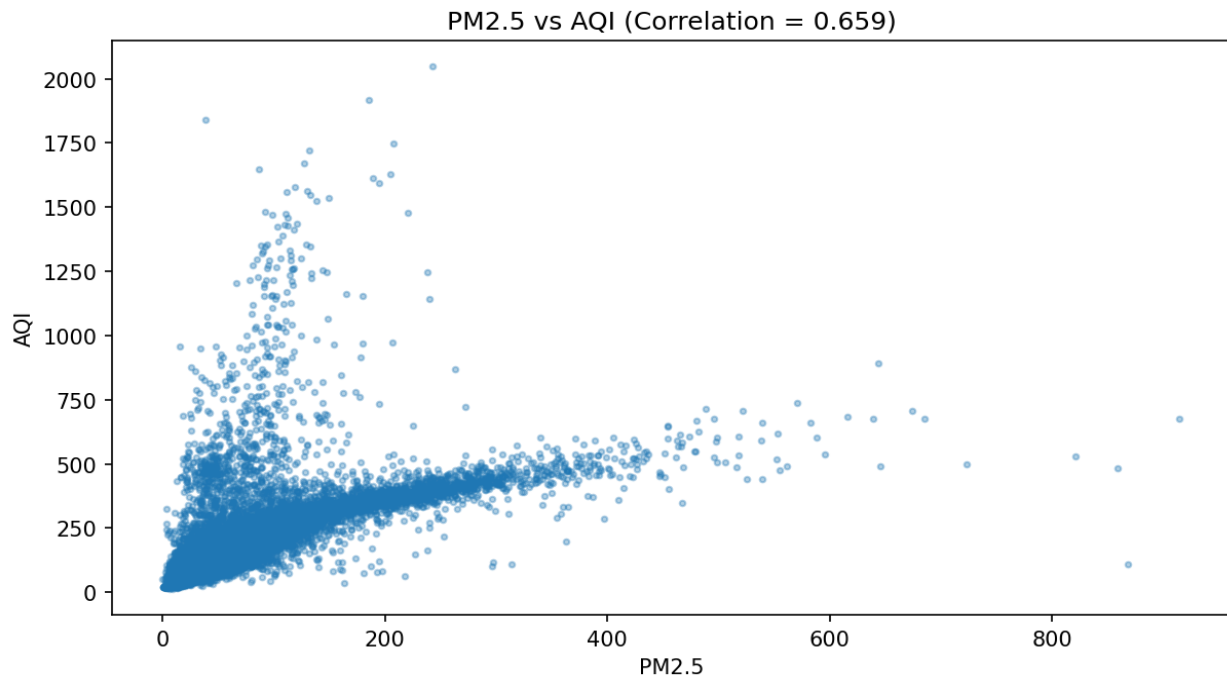
## 18. Analysis 6 – PM2.5 vs AQI

```
x = df[["PM2.5", "AQI"]].dropna()
```

```
plt.scatter(x["PM2.5"], x["AQI"])
plt.xlabel("PM2.5")
plt.ylabel("AQI")
plt.title("PM2.5 vs AQI")
plt.show()
```

```
print(x.corr())
```

### Visualization



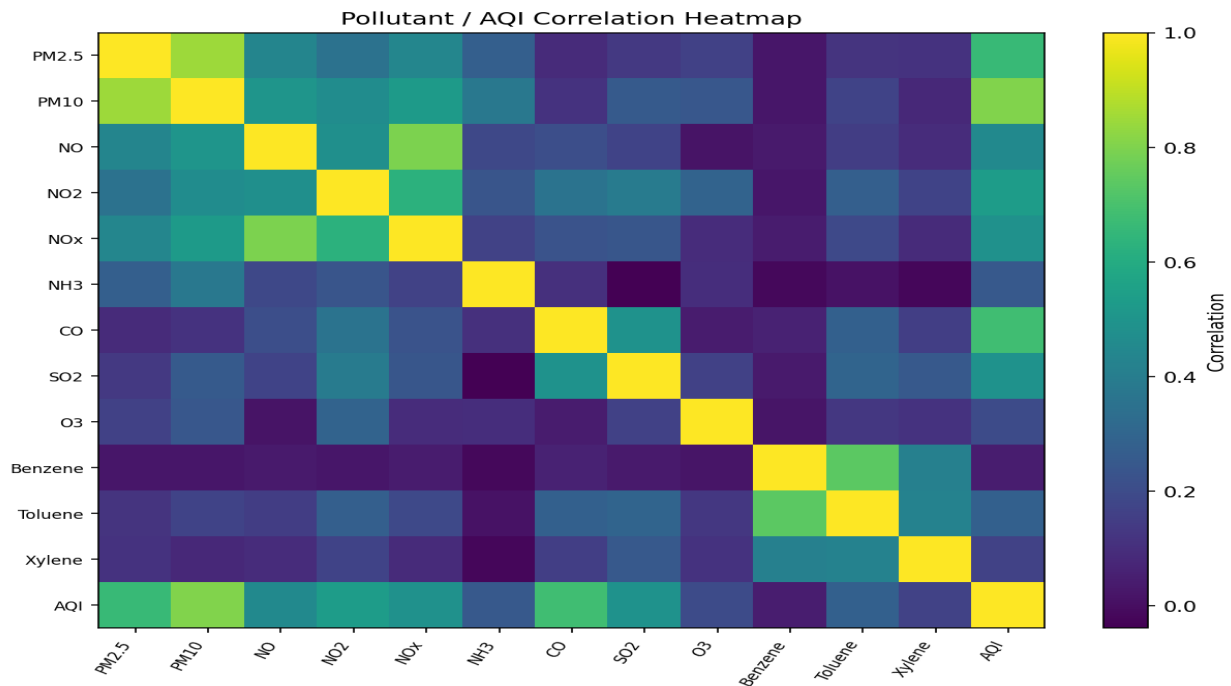
**Interpretation:** The scatter plot compares PM2.5 with AQI using rows where both values are available. The Pearson correlation is approximately **0.659**, indicating a positive relationship in this dataset. Higher PM2.5 values generally occur with higher AQI values, although the points are not perfectly aligned.

**PM2.5–AQI correlation:** 0.659181

## 19. Analysis 7 – Correlation Analysis

```
pollutants = ["PM2.5", "PM10", "NO", "NO2", "NOx", "NH3", "CO", "SO2", "O3", "Benzene", "Toluene", "Xylene"]  
  
corr = df[pollutants + ["AQI"]].corr()  
print(corr)  
  
plt.imshow(corr)  
plt.colorbar()  
plt.xticks(range(len(corr.columns)), corr.columns, rotation=60)  
plt.yticks(range(len(corr.index)), corr.index)  
plt.title("Pollutant / AQI Correlation Heatmap")  
plt.show()
```

### Visualization



**Interpretation:** The heatmap shows pairwise Pearson correlations among the pollutant variables and AQI. PM10 has the strongest positive correlation with AQI at 0.803, followed by CO at 0.683 and PM2.5 at 0.659. Correlation describes linear association and does not by itself prove causation.

## 20. Major Findings

**Finding 1:** Ahmedabad has the highest average AQI at **452.12**.

**Finding 2:** Aizawl has the lowest average AQI at **34.77** among cities with valid AQI observations.

**Finding 3:** Moderate AQI category occurs most frequently with **8,829** records.

**Finding 4:** PM10 has the highest average recorded pollutant value at **118.127**.

**Finding 5:** PM2.5 and AQI have a Pearson correlation of approximately **0.659**.

**Finding 6:** The dataset contains **29,531** rows, **16** columns and **26** cities.

**Finding 7:** There are **0** duplicate complete records in the supplied dataset.

**Finding 8:** The AQI values have a mean of **166.46**, median of **118.00**, minimum of **13** and maximum of **2049**.

## 21. Advantages

- Simple and easy-to-understand data analysis process.
- Uses Pandas for efficient tabular data handling.
- Visualizations make AQI and pollutant patterns easier to understand.
- City-wise grouping supports comparison between locations.
- Correlation analysis helps identify related variables.
- The analysis is based directly on the supplied dataset.

## 22. Future Scope

- Build an interactive dashboard for city and date selection.
- Apply machine-learning models for AQI prediction.
- Add more recent observations when available.
- Study seasonal and city-specific pollutant patterns in greater detail.
- Develop alert systems based on AQI categories.

## 23. Viva Questions

### 1. What is Air Quality Analytics?

**Answer:** It is the analysis of air-quality data to understand AQI and pollutant patterns.

### 2. What is AQI?

**Answer:** AQI is a numerical index used to represent air quality.

### 3. Why is EDA important?

**Answer:** EDA helps understand data structure, patterns, missing values and relationships.

### 4. Which Python library is used for data analysis?

**Answer:** Pandas is mainly used for data analysis.

### 5. Which library is used for visualization?

**Answer:** Matplotlib is used for creating graphs.

### 6. What is a histogram?

**Answer:** A histogram shows the frequency distribution of numerical values.

### 7. What is a scatter plot?

**Answer:** A scatter plot shows the relationship between two numerical variables.

### 8. What is correlation?

**Answer:** Correlation measures the strength and direction of a linear relationship between variables.

### 9. Which city has the highest average AQI in this dataset?

**Answer:** Ahmedabad, with an average AQI of 452.12.

### 10. What is the future scope of this project?

**Answer:** The project can be extended to AQI prediction, dashboards and real-time monitoring.

Note: Viva answers are intentionally short and based on the project and supplied dataset.

## 24. Result

Thus, the City Day air-quality dataset was successfully analyzed using Python, Pandas, Matplotlib, NumPy, and Exploratory Data Analysis techniques. Various statistical analyses and visualizations were performed to identify AQI trends, pollutant patterns, city-wise differences, and meaningful air-quality insights.

### Key numerical summary

| Item                       | Actual result            |
|----------------------------|--------------------------|
| Rows                       | 29,531                   |
| Columns                    | 16                       |
| Cities                     | 26                       |
| Date range                 | 01-01-2015 to 01-07-2020 |
| Valid AQI observations     | 24,850                   |
| Duplicate complete records | 0                        |
| Average AQI                | 166.46                   |
| Median AQI                 | 118.00                   |
| Minimum AQI                | 13                       |
| Maximum AQI                | 2049                     |
| Highest-average-AQI city   | Ahmedabad (452.12)       |
| Lowest-average-AQI city    | Aizawl (34.77)           |
| Most frequent AQI bucket   | Moderate (8,829)         |
| Highest average pollutant  | PM10 (118.127)           |
| PM2.5–AQI correlation      | 0.659181                 |

End of Mini Project Record